

Quantum Cryptography and Digital Evidence in Criminal Investigations

A Saudi Perspective, with a Reproducible Memory-Forensics Experiment

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Abstract

Quantum computing threatens the public-key cryptography that secures digital communications and, increasingly, the digital evidence on which criminal investigations depend. This paper examines that intersection from the perspective of Saudi Arabia, which modernized its evidentiary law through the Law of Evidence (Royal Decree M/43, in force July 2022) and which prosecutes computer crime under the Anti-Cyber Crime Law (2007). The paper reviews the relevant quantum and post-quantum cryptography, distinguishes the two effects that matter for evidence, namely the retrospective decryption of harvested data (“harvest now, decrypt later”) and the future forgeability of pre-quantum digital signatures, and analyses what each means for the collection, authentication and admissibility of digital evidence under Saudi law.

To ground the discussion in measurement rather than assertion, the paper reports a small, fully reproducible experiment on a standardized post-quantum algorithm. Using the liboqs implementation of ML-KEM-768, secret keys and shared secrets were generated and their persistence in process memory was examined before and after release. While the key material was in use, complete secret keys were recoverable from a memory snapshot; after the buffers were cleared with a secure-erase routine, no complete key and no shared secret remained, whereas ordinary deallocation left key fragments behind. This illustrates a central theme: quantum-resistant algorithms do not remove forensic opportunities, but they move them from cryptanalysis toward implementation and memory, which raises the importance of live acquisition. The paper closes with recommendations for Saudi investigators, forensic laboratories, courts and policymakers, and with a research agenda. It is a review and a proof-of-concept, not a study of real cases or a source of legal advice.

Keywords: quantum cryptography; post-quantum cryptography; digital forensics; digital evidence; Saudi Law of Evidence; Anti-Cyber Crime Law; memory forensics; ML-KEM; chain of custody

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1 Introduction

Digital evidence is now central to criminal investigations of almost every kind, from financial crime and fraud to violent crime, terrorism and offences committed wholly online. The integrity and authenticity of that evidence rest on cryptography: hash functions establish that a file has not changed, and digital signatures and secure channels establish who produced or sent something. Quantum computing threatens part of this foundation. A sufficiently large, error-corrected quantum computer running Shor’s algorithm would break the public-key cryptography, principally RSA and elliptic-curve cryptography, that underlies digital signatures and key exchange [1].

Saudi Arabia has modernized both the law and the technology around digital evidence. The Law of Evidence, promulgated by Royal Decree No. M/43 and in force from July 2022, for the first time codifies digital evidence as a recognized means of proof and defines its categories [12, 13]. Computer-related offences are prosecuted under the Anti-Cyber Crime Law of 2007 [16], and electronic signatures are governed by the Electronic Transactions Law [17]. At the same time, the Kingdom is investing in quantum technology: Aramco and Pasqal deployed Saudi Arabia’s first quantum computer, a 200-qubit research system, in Dhahran in November 2025 [11]. That machine cannot break cryptography, but it signals the direction of travel.

This paper asks how the arrival of quantum-resistant cryptography changes digital forensics, and what the consequences are for digital evidence under Saudi law. It is written for an audience that spans technical and criminal-justice practice, reflecting the author’s background in both.

1.1 Research questions

1. How does the transition to post-quantum cryptography change the technical methods available to digital-forensic examiners?
2. Where do forensic opportunities move as cryptanalytic attacks recede, and can this be demonstrated on a standardized post-quantum algorithm?
3. What do these changes mean for the collection, authentication and admissibility of digital evidence under Saudi Arabia’s Law of Evidence and related legislation?

1.2 Contributions and scope

The paper contributes (i) a focused review of the quantum threat as it bears on digital evidence, distinguishing confidentiality from authenticity effects; (ii) a small, fully reproducible experiment showing where key material can and cannot be recovered from memory for a standardized post-quantum KEM; and (iii) an analysis of the implications for Saudi evidentiary practice, with recommendations.

This is a single-author review and proof-of-concept. It does not analyse real criminal cases, does not include interviews or court data, and is expressly *not legal advice*; statements about Saudi law are drawn from the published legislation and secondary legal commentary and should be verified against the binding Arabic texts and qualified counsel. The experiment is a demonstration on the author’s own machine, not a validated forensic tool. These limits are revisited in Section 7.

2 Background: Quantum and Post-Quantum Cryptography

2.1 The quantum threat

Shor’s algorithm solves integer factorization and discrete logarithms in polynomial time on a quantum computer, breaking RSA, Diffie–Hellman, ECDSA and ECDH [1]. Grover’s algorithm gives only a quadratic speed-up against symmetric primitives [2], so AES-256 and SHA-2/3 remain practically secure with adequate parameters. No cryptographically relevant quantum computer (CRQC) exists today, and estimates of its arrival vary widely [3]; resource estimates for breaking RSA-2048 have fallen from about 20 million noisy qubits [4] to under one million [5], which is why standards bodies are acting now.

2.2 Standardized post-quantum cryptography

NIST published its first post-quantum standards in August 2024: ML-KEM for key encapsulation (FIPS 203), and ML-DSA and SLH-DSA for signatures (FIPS 204 and 205) [6–8]. Falcon is being standardized as FN-DSA (FIPS 206). These algorithms run on ordinary computers; their security rests on lattice or hash-function problems believed to resist quantum attack. This paper uses ML-KEM-768 for its experiment because it is the key-establishment parameter set most widely deployed in practice, including in the TLS hybrid used across the modern web.

2.3 Quantum key distribution

Quantum key distribution (QKD) is a distinct technology that uses quantum physics, rather than computational hardness, to establish keys, and detects eavesdropping through the disturbance that measurement causes [9]. QKD requires specialized hardware and is deployed only in niche settings; the NCA’s draft NCS-2 addresses it separately [20]. This paper treats QKD only briefly, because the cryptography that will actually appear in most evidence for the foreseeable future is software-based post-quantum cryptography, not QKD.

2.4 Two effects that matter for evidence

Two consequences of the quantum threat matter for digital evidence, and they have different timelines.

Retrospective decryption (confidentiality). Encrypted data recorded today can be stored and decrypted once a CRQC exists (“harvest now, decrypt later”). For investigators, this is double-edged: lawfully intercepted but currently unreadable material could become readable in the future, while sensitive case data and citizens’ data are themselves at risk if protected only by pre-quantum key exchange.

Retrospective forgeability (authenticity). A signature made today with RSA or ECDSA cannot be forged today, but once a CRQC exists an adversary could compute the corresponding private key and forge new signatures. This threatens the future evidentiary weight of pre-quantum digital signatures and time-stamps, unless their integrity is anchored by other means (for example, a secure timestamp or a hash committed to an append-only log before the CRQC era). Hash-based

integrity, by contrast, is only weakly affected by quantum attack and remains a dependable basis for evidence integrity.

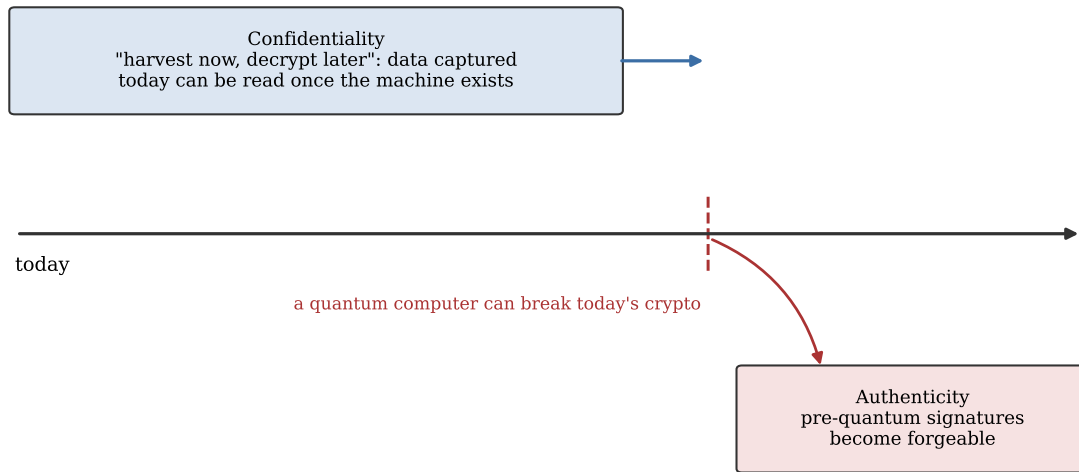


Figure 1: The two quantum effects that matter for digital evidence act on different timelines. Confidentiality is at risk now, because data captured today can be decrypted once a cryptographically relevant quantum computer (CRQC) exists. Authenticity is threatened only from the moment a CRQC exists, but that moment may fall within the lifetime for which a signature must remain trustworthy.

3 Impact on Digital Forensics

3.1 Where forensic leverage moves

Traditional access to encrypted evidence has relied on a mix of legal compulsion, implementation weaknesses and, occasionally, cryptanalysis. Post-quantum algorithms do not change the legal routes, and they close the (largely theoretical, for well-implemented modern crypto) cryptanalytic route further still. What they do is shift the practical leverage toward implementation and operational security: memory that still holds keys, artefacts left on disk, side channels in implementations, and metadata that survives regardless of payload encryption. This is not unique to post-quantum cryptography, but the transition is a natural moment to re-centre forensic method on these vectors.

Table 1: Where forensic access to protected data comes from, and how the move to post-quantum cryptography changes each route.

Route	Before (classical)	After (post-quantum)
Legal compulsion	Compel passwords or keys	Unchanged
Cryptanalysis	Rarely feasible for modern crypto	Closed further; not a realistic route
Implementation flaws	Sometimes exploitable	More important; new code, new bugs
Memory (keys in use)	Effective on live systems	More important; see experiment
Metadata	Survives encryption	Unchanged; still valuable

Two practical implications follow. First, *live acquisition grows in importance*: keys and plaintext are most reliably obtained while a system is running, before secure-erase routines wipe them. Second, *detection signatures must be updated*: memory-forensic tools recognize RSA and ECC key structures, but post-quantum keys have different sizes and layouts, so existing carving heuristics will miss them until updated.

3.2 A note on larger artefacts

Post-quantum keys and signatures are much larger than their classical counterparts (for example, an ML-DSA-44 signature is about 2.4 KB against 64 bytes for ECDSA, and an ML-KEM-768 key or ciphertext is roughly 1 KB against 32–65 bytes). Paradoxically, this can help forensic detection: larger, structured blobs are more distinctive in memory and storage once examiners know what to look for. The experiment below uses this fact.

4 Experiment: Recovering ML-KEM Keys from Memory

The claim that “forensic leverage moves to memory” is easy to assert and worth demonstrating. This section reports a small experiment that anyone can reproduce (Appendix A). It is a proof of concept about *where key material lives*, not a validated forensic procedure and not an attack on any third-party system: the program generates its own keys and inspects its own process memory.

4.1 Method

A C program using liboqs 0.16.0 creates two ML-KEM-768 key sets (labelled A and B). For each it generates a key pair, performs encapsulation and decapsulation, and confirms that both sides derive the same 32-byte shared secret. It writes the true secret keys (2,400 bytes each) and shared secrets to reference files. The program then pauses so a memory snapshot can be taken while all key material is in use (snapshot 1). Next it releases the buffers two different ways: key set A with the C library’s ordinary `free()`, which returns memory without clearing it, and key set B with liboqs’ `OQS_MEM_secure_free()`, which overwrites the memory before releasing it. It pauses again for a second snapshot (snapshot 2). A Python script then searches each snapshot for the reference secrets, counting both complete copies and the fraction of 32-byte blocks found anywhere (to detect partial or fragmented remnants).

The results reported here were produced on a Linux host, capturing the process’s writable memory through `/proc/<pid>/mem`. The procedure on the author’s macOS machine, using `gcore`, is given in Appendix A; independent runs will differ in detail, which is the point of reproduction.

4.2 Results

Table 2: Recovery of ML-KEM-768 secret material from process-memory snapshots. “Full copies” counts complete matches; “blocks found” is the share of the secret’s 32-byte blocks found anywhere in the snapshot.

Snapshot	Secret	Full copies	32-byte blocks found
1: in use	A secret key (2,400 B)	1	75/75 (100%)
1: in use	A shared secret (32 B)	2	1/1 (100%)
1: in use	B secret key (2,400 B)	1	75/75 (100%)
1: in use	B shared secret (32 B)	3	1/1 (100%)
2: after release	A secret key (freed normally)	0	74/75 (99%)
2: after release	A shared secret (freed normally)	0	0/1 (0%)
2: after release	B secret key (securely freed)	0	74/75 (99%)
2: after release	B shared secret (securely freed)	0	0/1 (0%)

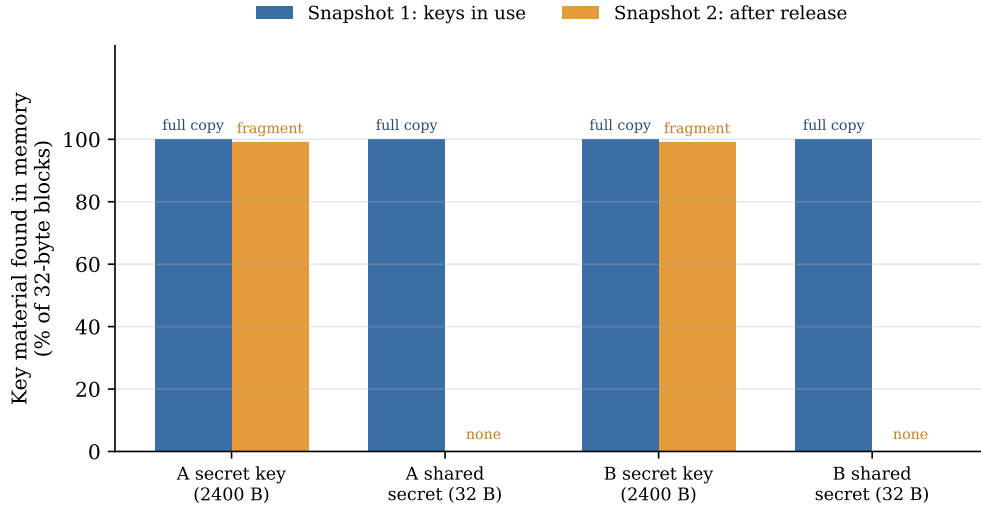


Figure 2: The same results as Table 2, shown as the share of each secret’s 32-byte blocks found in memory. While the keys were in use (snapshot 1) every secret was fully present. After the buffers were released (snapshot 2) no complete key survived and the shared secrets were gone; only isolated fragments of the large keys remained.

Three observations follow. First, while the keys were in use, every secret was fully present in memory, and the small shared secrets appeared in several copies, reflecting ordinary buffer duplication. Second, after release no *complete* secret key survived in either case, because the allocator reused or overwrote the large 2,400-byte regions once they were freed; only isolated fragments (about one 32-byte block) remained. Third, the 32-byte shared secrets vanished entirely after release under both methods. In this particular run, ordinary `free()` and secure erasure ended in a similar state for the large keys, because normal allocator behaviour happened to disturb the freed regions; the securely freed shared secrets were, as designed, wiped immediately.

4.3 Interpretation

The forensically important result is the contrast between the two snapshots, not the difference between the two free methods in this single run. *Keys are reliably recoverable only while in use.* Once a well-written application clears and releases key material, recovery of a complete key becomes unreliable at best. For investigators this argues strongly for live acquisition: capturing volatile memory from a running, unlocked system before keys are cleared is far more likely to yield usable key material than post-mortem analysis of a powered-down device. It also shows that whether secrets linger is an *implementation* property. Applications that use secure-erase routines leave less behind; those that do not may leave fragments that skilled examiners can sometimes exploit. The experiment does not attempt key reconstruction from fragments; it only measures presence.

A caution: this is one algorithm, one implementation, one platform and a single run. It demonstrates a mechanism; it does not quantify how often real applications leave keys in memory, which depends on their coding practices and is a question for future work on representative software.

5 Implications for Digital Evidence under Saudi Law

This section relates the technical picture to Saudi evidentiary practice. It summarizes publicly available descriptions of the relevant law and should be read as technical commentary, not legal advice; the binding texts are in Arabic and specific questions require qualified counsel.

5.1 The legal framework

The Law of Evidence (Royal Decree No. M/43), in force from July 2022, is the first codified law of evidence in the Kingdom and expressly recognizes digital evidence as a means of proof, alongside writing, testimony, presumption, inspection and expertise [12, 13]. Secondary commentary indicates that it defines digital evidence broadly, including digital records, documents, signatures, electronic communications and digital media, and makes such evidence admissible where standards of integrity and relevance are met [14, 15]. The law also envisages implementing regulations for digital-evidence procedures issued by the Minister of Justice in coordination with the Supreme Judicial Council [12]. Electronic signatures are recognized under the Electronic Transactions Law [17], and computer-related offences, including illegal interception and unauthorized access, are addressed by the Anti-Cyber Crime Law of 2007 [16]. Investigations are conducted within the framework of the Law of Criminal Procedure [18].

5.2 Authentication and integrity

Digital evidence must be shown to be authentic and unaltered. In practice this rests heavily on cryptographic hashing: an examiner records a hash of acquired data and can later demonstrate that the data is unchanged. The good news for Saudi practice is that this foundation is quantum-resistant: hash functions are only weakly affected by quantum attack, so hash-based integrity verification remains sound in the post-quantum era. Forensic laboratories should nonetheless ensure they use strong hash functions (SHA-256 or SHA-3 family) rather than legacy functions, and can add quantum-resistant digital signatures or timestamps to their evidence-handling records to protect the long-term authenticity of the records themselves.

5.3 Authenticity of signed evidence over time

Where the evidence *is* a digital signature, for example a signed contract, a signed log, or a code-signed artefact, the retrospective-forgeability effect matters. A signature valid today may lose evidentiary weight in future if the signing algorithm is quantum-vulnerable and a CRQC later emerges, because the possibility of forgery undermines the inference from signature to signer. Two mitigations are worth noting for Saudi practice: migrating high-value signing (for national PKI, government documents and long-lived records) to post-quantum or hybrid signatures, consistent with the NCA's National Cryptographic Standards [19, 20]; and preserving independent integrity anchors (secure timestamps, or hashes committed to append-only records) so that a signature's validity at the time of signing can still be established later.

5.4 Collection: the case for live acquisition

The experiment's practical lesson maps directly onto collection procedure. Because keys are reliably recoverable only while in use, and because strong encryption otherwise renders storage unreadable,

the value of capturing volatile memory from a live, unlocked device rises in the post-quantum era. This has procedural consequences under the Law of Criminal Procedure and any digital-evidence implementing regulations: search and seizure authorities and forensic standard operating procedures should support lawful live acquisition, with careful documentation, rather than defaulting to powering devices down. Live acquisition changes the evidence in subtle ways (a running system is not frozen), so chain-of-custody documentation must record exactly what was done, by whom and when.

5.5 Admissibility and the expert’s role

Post-quantum forensics is more complex than classical forensics, and courts will rely on expert testimony to explain it. For Saudi courts, three points follow. First, examiners should document methodology transparently, including tool versions and, where possible, reproducible steps, so that the reliability of a novel method can be assessed. Second, the field should invest in qualified experts and in shared, validated methodologies, so that expert evidence rests on accepted practice rather than individual assertion. Third, because much post-quantum forensic method targets implementation and memory rather than mathematics, expert explanations can often be framed in intuitive terms (“the key was still in the computer’s working memory”), which aids comprehension.

5.6 Cross-border cooperation

Cyber-enabled crime routinely crosses borders, and evidence often sits with foreign providers. The regional move toward post-quantum readiness, including the UAE’s National Encryption Policy [21] and Saudi Arabia’s NCS updates [20], will proceed at different speeds in different jurisdictions, which can create mismatches in what evidence is available and how it is protected. Harmonized approaches and mutual legal assistance remain important; this is a policy matter beyond the scope of this paper, noted here for completeness.

6 Recommendations

6.1 For investigators and forensic laboratories

1. Prioritize lawful live acquisition of volatile memory from running, unlocked devices, with rigorous documentation, before keys are cleared.
2. Use and record strong, quantum-resistant hash functions for evidence integrity, and consider post-quantum signatures or timestamps on evidence records.
3. Update or procure memory-forensic tooling that can recognize post-quantum key structures, whose sizes and layouts differ from RSA and ECC.
4. Build and document reproducible procedures so that novel methods can be explained and defended in court.

6.2 For courts and the judiciary

1. Support education for judges on cryptography and digital forensics at a conceptual level, focusing on integrity, authenticity and the meaning of a signature over time.
2. Encourage transparent, reproducible methodology as a factor in assessing the reliability of digital-forensic evidence.

6.3 For policymakers and standards bodies

1. Ensure that digital-evidence implementing regulations under the Law of Evidence accommodate live acquisition and the documentation it requires.
2. Align long-term signing for national PKI and government records with post-quantum or hybrid algorithms as the NCS is updated.
3. Support training and shared forensic methodologies, and regional cooperation on cross-border digital evidence.

7 Limitations

Proof of concept, not a tool. The experiment demonstrates a mechanism on one algorithm, one implementation and one platform, in a single run. It does not measure how often real applications leave keys in memory, and it does not reconstruct keys from fragments.

Not legal advice. Statements about Saudi law rely on the published legislation and secondary English-language commentary. The binding texts are in Arabic, implementing regulations continue to develop, and specific matters require qualified counsel.

No case data. The paper analyses no real investigations or court decisions and includes no interviews or survey data.

QKD out of scope. Quantum key distribution is treated only briefly, as it is not the cryptography most evidence will involve in the near term.

8 Conclusion and Future Work

Quantum-resistant cryptography does not end digital forensics; it relocates it. As cryptanalytic shortcuts recede, forensic leverage moves toward implementation, memory and metadata. The experiment reported here makes this concrete for a standardized algorithm: ML-KEM secret keys were fully recoverable from memory while in use, but not after the application cleared them, which is a direct argument for live acquisition. For Saudi Arabia, whose Law of Evidence now recognizes digital evidence and whose national cryptographic standards are being updated for the post-quantum era, the practical priorities are to strengthen live-acquisition capability and documentation, to rely on quantum-resistant hashing for integrity, to plan post-quantum protection for long-lived signed records, and to build expert capacity and reproducible methods that courts can trust.

Future work should measure key-in-memory behaviour across representative real applications and post-quantum libraries; test recovery of keys from partial memory fragments; extend the experiment to signature schemes and to embedded devices; and examine, with legal scholars, how the implementing regulations under the Law of Evidence should treat live acquisition and novel forensic methods.

Data and code availability

The experiment’s source code, analysis script and reproduction instructions are included with this paper’s source files (`kit/`).

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A Reproducing the Experiment

The code is in `kit/`: `kemmem.c` (generates keys and pauses for snapshots), `search_snapshot.py` (searches a snapshot), `snapshot_mac.sh` (captures a snapshot on macOS) and `README.md`.

Build against a local `liboqs` (0.16.0), reusing an existing build or making a library-only build:

```
cc -O2 -I ~/liboqs/build/include kemmem.c ~/liboqs/build/lib/liboqs.a -o kemmem
```

Run `./kemmem` in one terminal; it prints its PID and pauses. In a second terminal capture memory at each pause (macOS: `sudo ./snapshot_mac.sh <PID> snap1`, then `snap2`; Linux: read `/proc/<pid>/mem` over the writable regions in `/proc/<pid>/maps`). Analyse with:

```
python3 search_snapshot.py snap1.core  
python3 search_snapshot.py snap2.core
```

The reference secrets are written to `truth_*.bin` for this run only; delete them and the snapshots afterwards. Independent runs will not reproduce the exact counts in Table 2; the reproducible finding is the qualitative contrast between snapshot 1 (full keys present) and snapshot 2 (no complete key, shared secrets gone).

B Glossary

CRQC

Cryptographically relevant quantum computer.

Harvest now, decrypt later

Recording encrypted data now to decrypt once a CRQC exists.

KEM

Key-encapsulation mechanism, a public-key method for establishing a shared secret.

Live acquisition

Collecting data, especially volatile memory, from a running system.

ML-KEM

Module-Lattice-Based Key-Encapsulation Mechanism (FIPS 203), formerly Kyber.

QKD

Quantum key distribution: key establishment based on quantum physics.

Shared secret

The symmetric key agreed by a KEM, here 32 bytes for ML-KEM-768.

Secure erase

Overwriting memory before releasing it, so secrets do not linger.